Milestone 5

Model Evaluation & Analysis

Hindi → Haryanvi → Speech Pipeline

Team Contributions

Abhishek

- Evaluated LLaMA model
- Analyzed BLEU, chrF, Exact Match
- Investigated evaluation issues

Sanket Agrawal

- Compiled evaluation results
- Structured report & tables
- Organized outputs

Satyam Srivastava

- Evaluated TTS using .wav samples
- Performed human evaluation
- Created PPT & integrated results

Md Fazlur Rahman

- Conducted TTS error analysis
- Identified limitations
- Supported model analysis

Overview

- Text-to-Text: **LLaMA 3.1 (QLoRA)**
- Text-to-Speech: **Coqui VITS**
- End-to-End Dialect Pipeline

Evaluation Dataset

- Text Dataset: ~5000 samples
- Train: 4000 | Val: 500 | Test: 500
- TTS Dataset: ~5500 samples
- Filtered: ~2764 (~1.5 hours audio)

Preprocessing

- Duplicate & null removal
- Sentence length filtering
- Tokenization (LLM)
- Audio resampling (22050 Hz)
- LJSpeech format conversion

Evaluation Environment

- GPU: NVIDIA A100 (40GB)
- Python 3.12
- PyTorch 2.10
- Transformers 5.0
- BF16 Precision

Evaluation Metrics

- **Translation:** Exact Match, BLEU, chrF
- **TTS:** Mel Loss, KL Loss
- Generator & Discriminator Loss
- Duration Loss

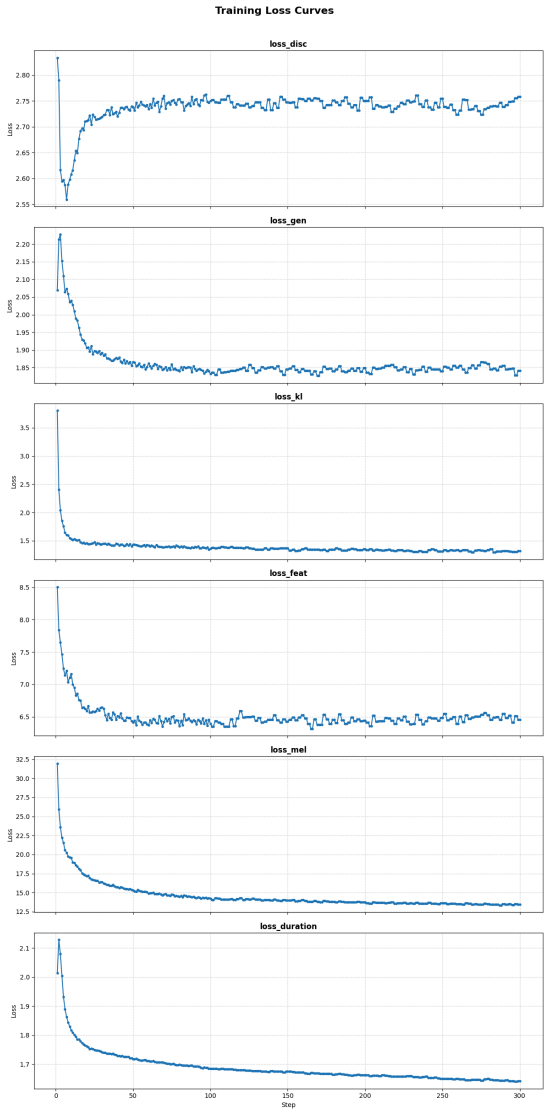
Best Model Results

Metric	Value
Eval Loss	0.00324
BLEU	71.76
chrF	77.65
Exact Match	50.42%

Experiment Comparison

Experiment	Eval Loss
baseline	0.00436
lower_lr_longer	**0.00324**
higher_rank	0.00427

TTS Training Curves



Qualitative Human Evaluation

- Evaluated on **11 inference samples**
- Includes short, medium, and long-form speech
- Samples range from **phrases** → **5-minute stories**
- Self-evaluation used (no MOS panel)

Inference Sample Coverage

Sample Type	Description
Short phrases	Idiomatic Haryanvi sentences
Conversational	Multi-line dialogues
Narrative	Story-based inputs
Long-form	~5 minute generated audio

Evaluation Observations

- **Prosodic Naturalness:** Strong rhythm and pacing
- **Hindi Bias:** Occasional Hindi words appear
- **Complex Sentences:** Stable without attention collapse
- **Short Sentences:** Best quality outputs

Performance Summary

Dimension	Assessment
Intelligibility	✅ High
Naturalness	✅ Strong
Hindi Bias	⚠️ Occasional
Long-form Stability	✅ Stable
Short Sentence Quality	✅ Excellent

Conclusion

Strong translation + stable TTS generation

End-to-end pipeline works effectively

High potential for low-resource dialect AI